

MATH 123 LECTURE NOTES

WEEK XII

ALTERNATING SERIES, ABSOLUTE AND CONDITIONAL CONVERGENCE

→ A series which the terms are alternately positive and negative is an alternating series. Here are three ^{examples} examples.

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots + \frac{(-1)^{n+1}}{n} + \dots \quad (1)$$

$$-2 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots + \frac{(-1)^n 4}{2^n} + \dots \quad (2)$$

$$1 - 2 + 3 - 4 + 5 - 6 + \dots + (-1)^{n+1} n + \dots \quad (3)$$

We see from these examples that the n th term of an alternating series is of the form

$$\underline{a_n = (-1)^{n+1} u_n \quad \text{or} \quad a_n = (-1)^n u_n}$$

where $u_n = |a_n|$ is a positive number

THM - THE ALTERNATING SERIES TEST (LEIBNIZ'S TEST)

The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \dots$$

converges if all three of the following conditions are satisfied

① The u_n 's are all positive

② The positive u_n 's are (eventually) nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .

③ $u_n \rightarrow 0$.

Ex 7 The following two series converge.

$$\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \rightarrow \text{Alternating harmonic series.}$$

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{2^n} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} + \frac{1}{256} + \dots$$

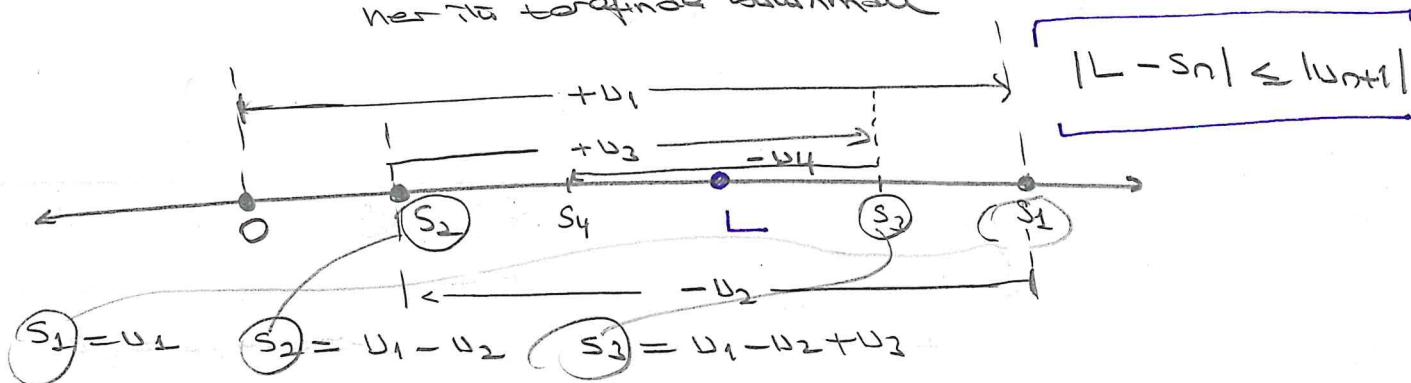
→ The partial sums of an alternating series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n$$

where $u_n \geq u_{n+1}$ for all $n \geq 1$ and $u_n \rightarrow 0$

(They straddle the limit L)

never too far from each other



THM The Alternating Series Estimation Theorem

If the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} u_n$ satisfies the three conditions.

- ① The u_n 's are all positive.
- ② The positive u_n 's are (eventually) nonincreasing $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
- ③ $u_n \rightarrow 0$

then for $n \geq N$

$$S_n = u_1 - u_2 + \dots + (-1)^{n+1} u_n$$

approximates the sum L of the series with an error whose absolute value is less than u_{n+1} , the absolute value of the first unused term. Furthermore, the sum L lies between any two successive partial sums S_n and S_{n+1}

and the remainder (2) $L - S_n$ has the same sign as the

Ex Let's try the preceding theorem on a series whose sum we know

$$\sum_{n=0}^{\infty} (-1)^n \frac{1}{2^n} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \frac{1}{64} - \frac{1}{128} + \frac{1}{256} - \dots$$

$S_8 = 0.6640625$

$$\frac{1}{1 - (-1/2)} = \frac{1}{3/2} = \frac{2}{3}$$

$$L - S_8 \leq a_9 = 1/256$$

The difference $L - S_8 = \frac{2}{3} - 0.6640625 = 0.002604166$ is positive and less than $1/256 = 0.00390625$

Absolute and Conditional Convergence

Definition A series $\sum a_n$ converges absolutely (is absolute convergent) if the corresponding series of absolute values, $\sum |a_n|$ converges.

Ex The geometric series

$$\sum_{n=0}^{\infty} (-1)^n \cdot \frac{1}{2^n} = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \dots$$

Converges absolutely because $\left| \frac{(-1)^n}{2^n} \right| = \left| \left(\frac{1}{2} \right)^n \right|$

$\sum_{n=0}^{\infty} \frac{1}{2^n} = 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots$ converges. The alternating

harmonic series $\sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n}$ does not converge

absolutely because the corresponding series of absolute

values $\sum_{n=1}^{\infty} \left| (-1)^n \cdot \frac{1}{n} \right| = \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent harmonic series.

(3)

Definition A series that converges but does not converge absolutely converges conditionally

- The alternating harmonic series converges conditionally

$$\underbrace{\sum (-1)^n \cdot \frac{1}{n}}_{\text{alternating harmonic series}} \quad \left. \begin{array}{l} \textcircled{1} u_n = 1/n > 0 \\ \textcircled{2} u_n \geq u_{n+1} \quad 1/n \geq 1/(n+1) \\ \textcircled{3} \lim_{n \rightarrow \infty} 1/n = 0 \end{array} \right\} \begin{array}{l} \text{Alternating} \\ \text{Series test} \\ \text{(A.S.T)} \end{array}$$

According to A.S.T $\sum_{n=1}^{\infty} (-1)^n \cdot 1/n$ is convergent.

However, $\sum_{n=1}^{\infty} |(-1)^n \cdot \frac{1}{n}| = \sum_{n=1}^{\infty} 1/n$ Harmonic series is

divergent. Therefore, alternating harmonic series converges conditionally.

THE ABSOLUTE CONVERGENCE TEST

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

Ex ⑦ $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{1}{n^2}$ check if it is absolute or conditionally convergent?

$$\sum_{n=1}^{\infty} |(-1)^{n+1} \cdot \frac{1}{n^2}| = \sum_{n=1}^{\infty} \frac{1}{n^2} \quad p=2 > 1 \Rightarrow \text{it converges.}$$

Therefore, $\sum_{n=1}^{\infty} (-1)^{n+1} \cdot \frac{1}{n^2}$ converges and it converges absolutely.

$$\textcircled{b} \sum_{n=1}^{\infty} \frac{\sin n}{n^2} = ?$$

$$\sum_{n=1}^{\infty} \frac{\sin n}{n^2} = \frac{\sin 1}{1^2} + \frac{\sin 2}{2^2} + \frac{\sin 3}{3^2} + \dots$$

$$\sum_{n=1}^{\infty} \left| \frac{\sin n}{n^2} \right| = \frac{|\sin 1|}{1} + \frac{|\sin 2|}{2} + \dots$$

$$|s_{nn}| \leq 1$$

$$\left| \frac{\sin \theta}{n_2} \right| \leq \frac{1}{n_2}$$

Since

$$\sum \frac{1}{n^2} \quad p=2 > 1$$

convergent. By using comparison let me say that

$\sum \left(\frac{\sin n}{n} \right)$ is also convergent. Therefore, $\sum \frac{\sin n}{n}$

converges absolutely.

(c) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$ check for convergence?

$$\frac{1}{c^p} = \frac{1}{c^p}$$

$$\sum \frac{1}{n^p} \Rightarrow \text{convergent for } p > 1$$

therefore, for $P \geq 1$ using comparison test.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \Rightarrow \text{absolute convergent.}$$

For $0 < p \leq 1$, $\sum \frac{1}{n^p}$ divergent.

Beweis

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^p}$$

$$V_0 = \frac{1}{14C}$$

⑤ 570

(2) $u_{n+1} \leq u_n \quad \frac{1}{(n+1)^p} \leq \frac{1}{n^p}$

③ $\lim_{p \rightarrow \infty} \frac{1}{p} = 0$.

Therefore, A.S.T gives
T doesn't. Therefore

931

$$\frac{(1) \times 1}{14}$$

no
duper
90

although $\sum \frac{1}{n^2}$ conditionally converges

POWER SERIES

Definition 1 A power series about $x=0$ is a series of the form

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \dots + c_n x^n + \dots \quad (1)$$

A power series about $x=a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \dots + c_n (x-a)^n + \dots \quad (2)$$

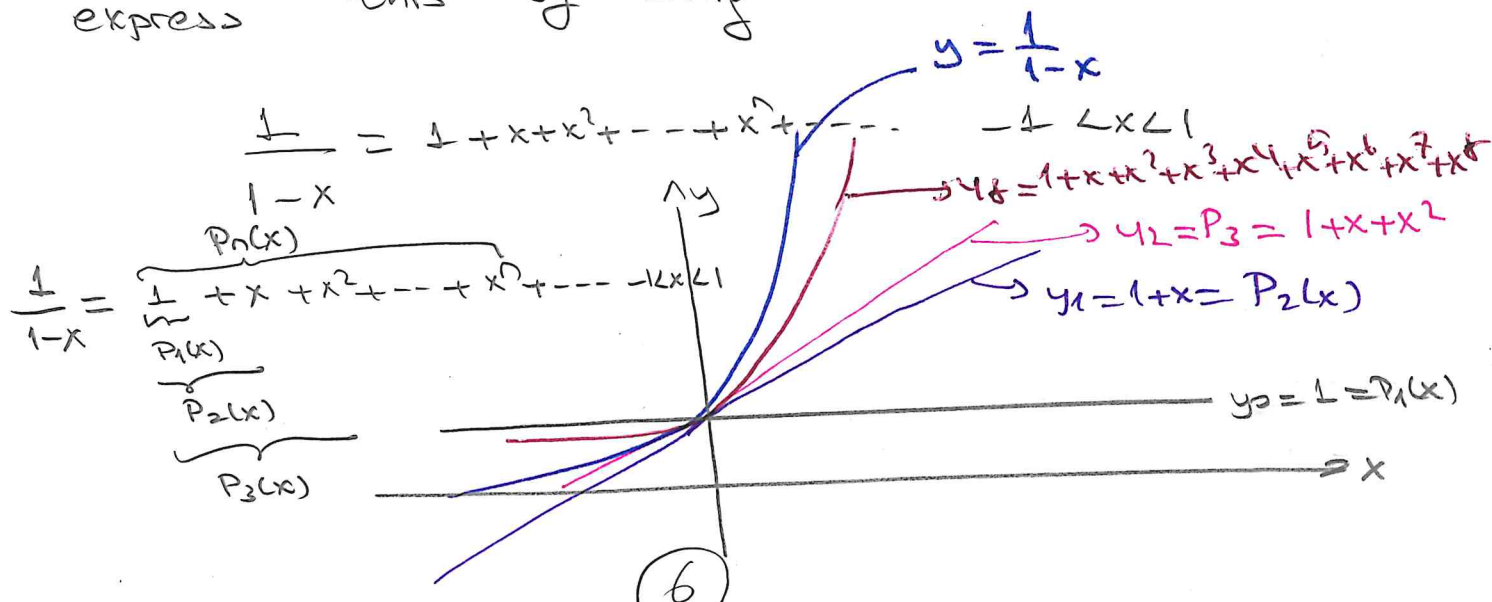
in which the center a and the coefficients $c_0, c_1, c_2, \dots$ are constants.

Ex 7 Taking all coefficients to be 1 in equation (1) gives geometric power series

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots + x^n + \dots \rightarrow \text{Geometric series with first term 1 and ratio } x.$$

It converges to $\frac{1}{1-x}$ for $|x| < 1$. We

express this by writing



Ex The power series

$$1 - \frac{1}{2}(x-2) + \frac{1}{4}(x-2)^2 + \dots + (-\frac{1}{2})^n (x-2)^n + \dots$$

corresponds $a=2$, $c_0=1$, $c_1=-1/2$, $c_2=1/4$, $\dots$, $c_n=(-1/2)^n$. This

is a geometric series with first term 1 and ratio

$$r = -\left(\frac{x-2}{2}\right)$$

$$\sum \underbrace{\left(-\frac{1}{2}\right)^n}_{c_n} \underbrace{(x-2)^n}_{(x-a)^n} = \sum \left(-\frac{x-2}{2}\right)^n$$

It converges to $\frac{1}{1-r} = \frac{1}{1 + \frac{x-2}{2}} = \frac{2}{x}$ for

$$\left|\frac{x-2}{2}\right| < 1 \quad \text{or} \quad 0 < x < 4.$$

So, we write

$$\frac{2}{x} = 1 - \frac{(x-2)}{2} + \frac{(x-2)^2}{4} + \frac{(x-2)^3}{8} + \dots + (-\frac{1}{2})^n (x-2)^n + \dots \quad 0 < x < 4$$

Ex For what values of x do the following series converge?

② $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$

Apply Ratio Test to the series $\sum |u_n|$ where u_n is the
 n th term of the power series in question

$$\left| (-1)^{n-1} \frac{x^n}{n} \right| = \left| \frac{x^n}{n} \right|$$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| = \frac{n}{n+1} |x| \quad \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{n}{n+1} |x| = |x|$$

The series converges absolutely for $|x| < 1$

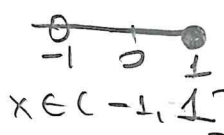
It diverges if $|x| > 1$ because the n th term doesn't converge to 0.

At $x=1$, we get alternating harmonic series. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$
which converges by A.S.T. $1 - 1/2 + 1/3 - 1/4 + \dots$

At $x=-1$, we get $\sum_{n=1}^{\infty} \frac{(-1)^{2n-1}}{n} = -1 - 1/2 - 1/3 - 1/4 - \dots$

the negative harmonic series. diverges. so we can say

$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$ converges for $-1 < x \leq 1$ and

diverges everywhere convergence interval 

$$(5) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{2n-1}}{2n-1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

$$\left| (-1)^{n-1} \frac{x^{2n-1}}{2n-1} \right| = \left| \frac{x^{2n-1}}{2n-1} \right| = |u_n|$$

$$\text{Ratio test } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2n+1}}{2n+1} \cdot \frac{2n-1}{x^{2n-1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| x^2 \frac{2n-1}{2n+1} \right| = x^2$$

Series absolutely convergent for $x^2 < 1$

If $x^2 > 1$, it diverges

Check Boundary points $x=1, x=-1$

$x=1$ $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1}$ Use A.S.T

$u_n = \frac{1}{2n-1}$

① $u_n > 0$

② $u_{n+1} = \frac{1}{2n+1} < u_n = \frac{1}{2n-1}$

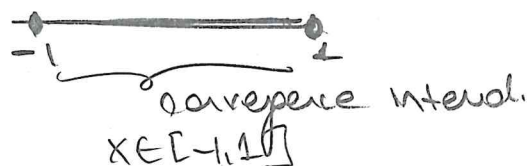
So $x=1$ we have a convergent Series. $\lim_{n \rightarrow \infty} u_n = 0$

$x=-1$ $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{(-1)^{2n-1}}{2n-1} = \sum_{n=1}^{\infty} \frac{(-1)^{3n-2}}{2n-1} = \sum_{n=1}^{\infty} \frac{(-1)^n}{2n-1}$

$(-1)^{3n-2} = \underbrace{(-1)^{2n-2}}_{\text{even}} \cdot (-1)^n = (-1)^n$

A.S.T $u_n = \frac{1}{2n-1}$ same as before it is also convergent

So we have convergence sequence for $-1 \leq x \leq 1$ and diverges elsewhere



③ $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

Apply Ratio test to the series $\sum |u_n|$ where u_n is the n th term of the power series

$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \frac{|x|}{n+1}$

$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 < 1$ for all values of x $0 < 1$

Therefore this series converges absolutely for all $x \in \mathbb{R}$



(d) $\sum_{n=0}^{\infty} n! x^n = ?$

$$\left| \frac{u_{n+1}}{u_n} \right| = \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| = |(n+1)x|$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} (n+1)|x| = \infty \text{ unless } x=0$$

The series diverges all values of x except $x=0$ 

THM The Convergence Theorem for Power Series

If the power series $\sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$ converges at $x=c \neq 0$, then it converges absolutely for all x with $|x| < |c|$. If the series diverges at $x=d$, then it diverges for all x with $|x| > |d|$.

The Radius of Convergence of a Power Series

COROLLARY The convergence of the series $\sum c_n (x-a)^n$ is described by the following three cases

① There is a positive number R such that the series diverges for x with $|x-a| > R$ but converges ^{absolutely} for x with $|x-a| < R$. The series may or may not converge at either of the endpoints $x=a-R$ and $x=a+R$.

② The series converges absolutely for every x ($R=\infty$)

③ The series converges at $x=a$ and diverges everywhere else ($R=0$)

(See result on page 12)

NOT Suppose $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ $R = \frac{1}{L}$ if $L=0$ $R=\infty$
Then the power series

Ex Determine the centre, radius and interval of convergence

a $\sum_{n=0}^{\infty} \frac{(2x+5)^n}{(n^2+1)3^n} = ?$

b $\sum_{n=0}^{\infty} \frac{n(x+3)^n}{5^n}$

c $\sum_{n=0}^{\infty} x^n$

$$\sum_{n=0}^{\infty} \underbrace{\left(\frac{2}{3}\right)^n \frac{1}{n^2+1} \left(x+\frac{5}{2}\right)^n}_{a_n} = \sum_{n=0}^{\infty} a_n$$

Ratio Test

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n^2+1)3^n}{(n^2+2n+2)3^{n+1}} \quad |2x+5| \rightarrow \frac{1}{3} |2x+5|$$

$$\Rightarrow -3 < 2x+5 < 3$$

$$-4 < x < -1 \Rightarrow R = -\frac{1-(-4)}{2} = \frac{3}{2} \text{ and the series converges}$$

absolutely on $(-4, -1)$; diverges if $x > -1$ or $x < -4$

$x = -1$ $\Rightarrow \sum \frac{1}{n^2+1}$ Converges by comparison test

$$\frac{1}{n^2+1} < \frac{1}{n^2} \quad p=2 > 1 \text{ converges} \quad \sum \frac{1}{n^2}$$

$x = -4$ $\Rightarrow \sum_{n=0}^{\infty} \frac{(-1)^n}{n^2+1}$ converges A.S.T

① $u_n = \frac{1}{n^2+1} > 0$

② $u_{n+1} < u_n \quad \frac{1}{(n+1)^2+1} < \frac{1}{n^2+1}$

③ $\lim_{n \rightarrow \infty} \frac{1}{n^2+1} = 0$

So convergence interval

$$x \in [-1, -4]$$

Ratio Test $\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)(x+3)^{n+1}}{5^{n+1}} \cdot \frac{5^n}{n(x+3)^n} \right|$

$= \lim_{n \rightarrow \infty} \frac{|x+3|}{5} \left(\frac{n+1}{n} \right) = \frac{|x+3|}{5} < 1$

$|x+3| < 5 \Rightarrow -8 < x < 2$ Radius $\frac{2 - (-8)}{2} = 5$

or we can use $L = \frac{1}{5}$ Radius $= \frac{1}{L} = 5$.

$x = -8$ $\sum_{n=0}^{\infty} (-1)^n \cdot n$ it is divergent because

n -th term test $\lim_{n \rightarrow \infty} (-1)^n n \neq 0$ divergent

$x = 2$ $\sum_{n=0}^{\infty} n$ $\lim_{n \rightarrow \infty} n = \infty$ divergent

Interval of convergence $x \in (-8, 2)$

Radius $R = 5$

(c) $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{x^n} \right| = |x|$ $|x| < 1$ $-1 < x < 1$ Radius $R = 1$

$x = -1$ $\sum_{n=0}^{\infty} (-1)^n$ divergent from

$\lim_{n \rightarrow \infty} (-1)^n \neq 0$ n th term test

$x = 1$ $\sum_{n=0}^{\infty} 1$ divergent

$\lim_{n \rightarrow \infty} 1 \neq 0$

Absolutely convergent in $-1 < x < 1$

Result Consider series $\sum_{n=0}^{\infty} a_n (x-a)^n$

$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1} (x-a)^{n+1}}{a_n (x-a)^n} \right| = \underbrace{\left(\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \right)}_L |x-a|$

$\rho < 1$ $\sum a_n (x-a)^n$ converges absolutely

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| |x-a| < 1$ $|x-a| < \frac{1}{\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|} = R$ $\rightarrow$ Radius of convergence

If $L = 0$ $R = \infty$ converges for all x

OPERATIONS ON POWER SERIES

THEOREM The Term by Term Differentiation Theorem

If $\sum c_n (x-a)^n$ has radius of convergence $R > 0$, it defines a function

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \text{ on the interval } a-R < x < a+R$$

This function f has derivatives of all orders inside the interval, and we obtain the derivatives by differentiating the original series term by term.

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1}$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n (x-a)^{n-2}$$

and so on. Each of these derived series converges at every point of the interval $a-R < x < a+R$.

THEOREM The Term by Term Integration Theorem

Suppose that $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ converges for $a-R < x < a+R$ ($R > 0$). Then $\sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}$ converges

$$a-R < x < a+R \text{ and } \int f(x) dx = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1} + C.$$

for $a-R < x < a+R$.

Ex 7 Find power series representation for the following functions

(a) $\frac{1}{(1-x)^2}$

(b) $\frac{1}{(1-x)^3}$

(c) $\ln(1+x)$

Starting with the following geometric series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \quad (-1 < x < 1)$$

Now, we can use differentiation, integration and substitution.

(a) Derivative $\left(\frac{1}{1-x}\right)' = \sum_{n=0}^{\infty} (x^n)'$

$$\frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + 4x^3 + \dots \quad (-1 < x < 1)$$

changes because for $n=0$
 $n x^{n-1} = 0$

(b) Take two times derivative to

$$\frac{2}{(1-x)^3} = \sum_{n=2}^{\infty} n(n-1) x^{n-2} = (1 \times 2) + (2 \times 3)x + (3 \times 4)x^2 + \dots$$

Now divide by 2.

$$\frac{1}{(1-x)^3} = \sum_{n=2}^{\infty} \frac{n(n-1)}{2} x^{n-2} = 1 + 3x + 6x^2 + 10x^3 + \dots \quad (-1 < x < 1)$$

(c) Substitute t instead of x

$$\frac{1}{1+t} = \sum_{n=0}^{\infty} (-1)^n t^n = 1 - t + t^2 - t^3 + t^4 - \dots$$

Integrate from 0 to x , where $|x| < 1$

$$\int_0^x \frac{1}{1+t} dt = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^n dt$$

$$\ln|1+t| \Big|_0^x = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{t^{n+1}}{n+1} \Big|_0^x$$

$$\ln|1+x| = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{n+1}}{n+1} \quad -1 < x < 1$$

$$\ln|1+x| = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} \quad (14)$$

Ex 7 Use geometric series of the previous example to find a power series representation of $\arctan x$

Substitute $-t^2$ instead of x in the geometric series.

Since $0 \leq t^2 < 1$ whenever $-1 < t < 1$ we obtain

$$\frac{1}{1+t^2} = \sum_{n=0}^{\infty} (-t^2)^n = \sum_{n=0}^{\infty} (-1)^n \cdot t^{2n} = 1 - t^2 + t^4 - t^6 + t^8 - \dots \quad (-1 < t < 1)$$

Now integrate from 0 to x , whenever $|x| < 1$.

$$\int_0^x \frac{1}{1+t^2} dt = \sum_{n=0}^{\infty} \int_0^x (-1)^n \cdot t^{2n} dt$$

$$\arctan x - \underbrace{\arctan 0}_0 = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad -1 < x < 1$$

$$\arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

Ex 7 Find a series representation of $f(x) = \frac{1}{2+x}$ in powers of $x-1$. What is the interval of convergence of the series.

Let $t = x-1$ so $x = t+1$. We have

$$\begin{aligned} \frac{1}{2+x} &= \frac{1}{3+t} = \frac{1}{3} \cdot \frac{1}{1+\frac{t}{3}} = \frac{1}{3} \sum_{n=0}^{\infty} (-1)^n \left(\frac{t}{3}\right)^n \quad -1 < t/3 < 1 \\ &= \frac{1}{3} \left[1 - \frac{t}{3} + \frac{t^2}{3^2} - \frac{t^3}{3^3} + \dots \right] \\ &\quad -1 < t/3 < 1 \end{aligned}$$

$$\begin{aligned} \frac{1}{2+x} &= \sum_{n=0}^{\infty} (-1)^n \frac{t^n}{3^{n+1}} = \sum_{n=0}^{\infty} (-1)^n \frac{(x-1)^n}{3^{n+1}} \quad \begin{aligned} &-3 < t < 3 \\ &-3 < x-1 < 3 \\ &-2 < x < 4 \end{aligned} \end{aligned}$$

TAYLOR AND MACLAURIN SERIES

THEOREM Suppose the series

$$f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \dots$$

converges to $f(x)$ for $c-\rho < x < c+\rho$ where $\rho > 0$.

Then, $a_k = \frac{f^{(k)}(c)}{k!}$ for $k=0, 1, 2, 3, \dots$

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \dots$$

$$f'(x) = \sum_{n=1}^{\infty} n a_n (x-c)^{n-1} = a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \dots$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) a_n (x-c)^{n-2} = 2a_2 + 6a_3(x-c) + 12a_4(x-c)^2 + \dots$$

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1) a_n (x-c)^{n-k}$$

$$\left(\frac{e^x}{2+x} \right)_{x=1} = \sum_{n=0}^{\infty} \frac{(-1)^n (x+1)^n}{3^{n+1}}$$

$$f(x) = \frac{1}{2+x} \quad f^{(30)}(1) = ?$$

$$a_{30} = \frac{f^{(30)}(1)}{30!}$$

$$= k! a_k + \frac{(k+1)!}{1!} a_{k+1}(x-c) + \frac{(k+2)!}{2!} a_{k+2}(x-c)^2 + \dots$$

$$a_n = \frac{(-1)^n}{3^{n+1}} = \frac{f^{(n)}(1)}{3^{n+1}}$$

$$f^{(30)}(1) = 30! a_{30}$$

$$a_{30} = \frac{(-1)^{30}}{3^{31}}$$

Taylor and Maclaurin Series

If $f(x)$ has derivatives of all orders at $x=c$ (i.e. $\frac{f^{(k)}(c)}{k!}$)

If $f^{(k)}(c)$ exists for $k=0, 1, 2, 3, \dots$ then the series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f^{(3)}(c)}{3!}(x-c)^3 + \dots$$

is called Taylor series of f about c (or Taylor series

of f in powers of $x-c$). If $c=0$, the term

Maclaurin Series is usually used in place of Taylor series

Analytic Function

A function f is analytic at c if f has a Taylor series at c and that series converges to $f(x)$ in an open interval containing c . If f is analytic at each point of an open interval, then we say that it is analytic on that interval.

Ex: Find the Taylor series for e^x about c . Where does the series converge to e^x ? Where is e^x analytic? What is the Maclaurin series for e^x ?

Solution Let $f(x) = e^x$. $f'(x) = e^x$ $f''(x) = e^x \dots f^{(n)}(x) = e^x$

The Taylor series of e^x about $x=c$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = \sum_{n=0}^{\infty} \frac{e^c}{n!} (x-c)^n = e^c + e^c(x-c) + \frac{e^c}{2!} (x-c)^2 + \dots$$

$$\text{Because } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\frac{e^c}{(n+1)!} (x-c)^{n+1}}{\frac{e^c}{n!} (x-c)^n} \right| = \lim_{n \rightarrow \infty} \frac{1}{n+1} |x-c| = 0 < 1$$

this series converges for all $x \in \mathbb{R}$ $R = \infty$.

Suppose the sum is $p(x)$

$$g(x) = \sum_{n=0}^{\infty} \frac{e^c}{n!} (x-c)^n = e^c + e^c(x-c) + \frac{e^c}{2!} (x-c)^2 + \frac{e^c}{3!} (x-c)^3 + \dots$$

$$\text{Then, } g'(x) = e^c + \frac{e^c}{2!} 2(x-c) + \frac{e^c}{3!} 3(x-c)^2 + \dots = p(x)$$

Since $p'(x) = p(x)$. It follows that $p(x) = Ke^x$ for some real number K . But $e^c = p(c) = Ke^c$ implies $K=1$,
∴ $e^c = p(x) = e^x$

The Taylor series for e^x about $x=c$ converges to e^x for every real number x :

$$e^x = \sum_{n=0}^{\infty} \frac{e^c}{n!} (x-c)^n$$

Setting $c=0$, we obtain the Maclaurin series for e^x

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{for } x \in \mathbb{R}$$

Ex 7 Find Maclaurin series for a) $\sin x$ and b) $\cos x$.

Where does each series converge

Solution Let $f(x) = \sin x$ Then $f(0) = 0$

$$f'(x) = \cos x \quad f'(0) = 1 \quad f'''(x) = -\cos x \quad f'''(0) = -1$$

$$f''(x) = -\sin x \quad f''(0) = 0 \quad f^{(4)}(x) = \sin x \quad f^{(4)}(0) = 0$$

Thus, the Maclaurin series

$$\begin{aligned} g(x) &= 0 + x + 0 - \frac{x^3}{3!} + 0 + \frac{x^5}{5!} + 0 - \dots \\ &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \end{aligned}$$

The series $g(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$ converges everywhere because

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1}}{(2n+3)!} x^{2n+3}}{\frac{(-1)^n}{(2n+1)!} x^{2n+1}} \right| = \lim_{n \rightarrow \infty} \frac{x^2}{(2n+3)(2n+2)} = 0 < 1$$

The radius of convergence is $R = \infty$

Series is convergent for all $x \in \mathbb{R}$

Let's show $p(x) = \sin x$

$$p'(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$p''(x) = -x + \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} - \dots = -p(x)$$

Thus $p(x)$ satisfies the differential equation $p''(x) + p(x) = 0$ where general solution is $p(x) = A \cos x + B \sin x$. Observe, from the series that $p(0) = 0$ and $p'(0) = 1$. This gives that

$$p(x) = \sin x \quad \text{and} \quad p'(x) = \cos x \quad \text{for all } x$$

Finally, we get

$$\left. \begin{aligned} \sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad (\forall x \in \mathbb{R}) \\ \cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \quad (\forall x \in \mathbb{R}) \end{aligned} \right\}$$

Some Maclaurin Series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots \quad (-1 < x < 1)$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1) x^n = 1 + 2x + 3x^2 + 4x^3 + \dots \quad (-1 < x < 1)$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad (-1 < x \leq 1)$$

$$\arctan x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)} x^{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad (-1 \leq x \leq 1)$$

$$e^{-x} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^n = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \quad (\forall x \in \mathbb{R})$$

$$\sinh x = \frac{e^x - e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \quad (\forall x \in \mathbb{R})$$

$$\cosh x = \frac{e^x + e^{-x}}{2} = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \quad (\forall x \in \mathbb{R})$$

Ex 7 Obtain Maclaurin series for the following functions

(a) $e^{-x^2/3}$

(b) $\frac{\sin(x^2)}{x}$

(c) $\sin^2 x$

(a) Substitute $-x^2/3$ for x in the Maclaurin series for e^x

$$e^{-x^2/3} = 1 - \frac{x^2}{3} + \frac{1}{2!} \left(\frac{x^2}{3}\right)^2 - \frac{1}{3!} \left(\frac{x^2}{3}\right)^3 + \dots$$

$$e^{-x^2/3} = \sum_{n=0}^{\infty} \frac{(-x^2/3)^n}{n!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{3^n \cdot n!}$$

(b) For all $x \neq 0$ we have

$$\frac{\sin(x^2)}{x} = \frac{1}{x} \left(x^2 - \frac{(x^2)^3}{3!} + \frac{(x^2)^5}{5!} - \dots \right)$$

$$= \frac{1}{x} \sum_{n=0}^{\infty} \frac{(x^2)^{2n+1} (-1)^n}{(2n+1)!}$$

$$= \frac{1}{x} \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+1}}{(2n+1)!}$$

Note that $f(x) = \frac{\sin(x^2)}{x}$ is not defined at $x=0$

but has a limit at $x=0$ which is 0.

$$\left(\lim_{x \rightarrow 0} \frac{\sin(x^2)}{x} \stackrel{(0/0)}{=} \lim_{x \rightarrow 0} \frac{\cos(x^2) \cdot 2x}{1} = 0 \right)$$

If we define $f(0) = 0$ (the continuous extension of $f(x)$ to $x=0$) then the series converges to $f(x)$ for all x values.

$$\textcircled{c} \sin^2 x = \frac{1 - \cos 2x}{2} = \frac{1}{2} - \frac{1}{2} \left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} - \dots \right)$$

$$= \frac{1}{2} \left(\frac{(2x)^2}{2!} - \frac{(2x)^4}{4!} + \frac{(2x)^6}{6!} - \dots \right)$$

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} 2^{2n+1}}{(2n+2)!} x^{2n+2} \quad (\forall x \in \mathbb{R})$$

Ex 7 Find the Taylor series for $f(x) = \ln x$ in powers of $x-2$. where does the series converge to $\ln x$?

$$\ln x = \ln(2 + (x-2)) = \ln\left(2 \left(1 + \frac{x-2}{2}\right)\right)$$

$$= \ln 2 + \ln(1+t)$$

we use Maclaurin series for $\ln(1+t)$

$$\ln x = \ln 2 + \ln(1+t)$$

$$= \ln 2 + t - \frac{t^2}{2} + \frac{t^3}{3} - \frac{t^4}{4} + \dots$$

$$= \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} t^n$$

$$= \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \left(\frac{x-2}{2}\right)^n = \ln 2 + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \frac{(x-2)^n}{2^n}$$

Since series of $\ln(1+t)$ is valid for $-1 < t \leq 1$,
this series for $\ln x$ is valid for $-1 < \frac{x-2}{2} \leq 1$

that is for $0 < x \leq 4$

Under what circumstances is a function equal to the sum of its Taylor series? In other words, if f has derivatives of all orders, when is it true that

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n \quad ?$$

$$T_n(x) = \sum_{i=0}^n \frac{f^{(i)}(a)}{i!} (x-a)^i$$

$$= f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!} (x-a)^n$$

The n th degree Taylor polynomial of f at a

$f(x)$ is the sum of its Taylor series if

$$f(x) = \lim_{n \rightarrow \infty} T_n(x)$$

$$R_n(x) = f(x) - T_n(x) \longrightarrow f(x) = \underbrace{T_n(x)}_{\text{Taylor polynomial}} + \underbrace{R_n(x)}_{\text{remainder}}$$

Thm If $f(x) = T_n(x) + R_n(x)$ where T_n is the n th-degree Taylor polynomial of f at a and

$$\lim_{n \rightarrow \infty} R_n(x) = 0$$

for $|x-a| < R$ then f is equal to the sum of its Taylor series on the interval $|x-a| < R$.

In trying to show that $\lim_{n \rightarrow \infty} R_n(x) = 0$ for a specific function f , we usually use the following fact.

TAYLOR'S INEQUALITY

If $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$, then the remainder $R_n(x)$ of the Taylor series satisfies the following inequality

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \text{ for } |x-a| \leq d$$

Ex 7 Prove that e^x is equal to the sum of its Maclaurin series

If $f(x) = e^x$, then $f^{(n+1)}(x) = e^x$ for all n . If d is any positive number and $|x| \leq d$, then $|f^{(n+1)}(x)| = e^x \leq e^d$. So, Taylor's Inequality with $a=0$ and $M=e^d$.

$$|R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1} \text{ for } |x| \leq d.$$

Notice that $M=e^d$ works for every value of n .

$$\lim_{n \rightarrow \infty} \frac{e^d}{(n+1)!} |x|^{n+1} = e^d \lim_{n \rightarrow \infty} \underbrace{\frac{|x|^{n+1}}{(n+1)!}}_0 = 0$$

$$\text{because } \lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

By the sandwich theorem $0 \leq |R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1}$

$$\lim_{n \rightarrow \infty} |R_n(x)| = 0 \Rightarrow \lim_{n \rightarrow \infty} R_n(x) = 0$$

Then, we get

$$(2) \quad e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \text{ for all } x \in \mathbb{R}$$

Ex 7 Find the Maclaurin series for $\sin x$ and prove that it represents $\sin x$ for all x

$$\begin{aligned} f(x) &= \sin x \Rightarrow f(0) = 0 & f''(x) &= -\sin x \Rightarrow f''(0) = 0 \\ f'(x) &= \cos x \Rightarrow f'(0) = 1 & f'''(x) &= -\cos x \Rightarrow f'''(0) = -1 \\ f^{(4)}(x) &= \sin x \Rightarrow f^{(4)}(0) = 0 \end{aligned}$$

$$f(0) + \frac{f'(0)}{1!} (x-0) + \frac{f''(0)}{2!} (x-0)^2 + \frac{f'''(0)}{3!} (x-0)^3 =$$

$$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Since $f^{(n+1)}(x)$ is $\pm \sin x$ or $\pm \cos x$, we know that $|f^{(n+1)}(x)| \leq 1$ for all x so we can take $M=1$

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x^{n+1}| = \frac{|x^{n+1}|}{(n+1)!} \xrightarrow{n \rightarrow \infty} 0$$

By squeeze theorem $0 \leq |R_n(x)| \leq \frac{M}{(n+1)!} |x^{n+1}|$

$|R_n(x)| \rightarrow 0$. It follows that $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Ex Find the Maclaurin series for $f(x) = (1+x)^k$ where k is any real number

$$f(x) = (1+x)^k$$

$$f(0) = 1$$

$$f'(x) = k(1+x)^{k-1}$$

$$f'(0) = k$$

$$f''(x) = k(k-1)(1+x)^{k-2}$$

$$f''(0) = k(k-1)$$

$$f'''(x) = k(k-1)(k-2)(1+x)^{k-3}$$

$$f'''(0) = k(k-1)(k-2)$$

$$f^{(n)}(x) = k(k-1)\dots(k-n+1)(1+x)^{k-n} \quad f^{(n)}(0) = k(k-1)\dots(k-n+1)$$

Therefore the Maclaurin series of $f(x) = (1+x)^k$ is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{k(k-1)\dots(k-n+1)}{n!} x^n$$

These series are called the binomial series. If its n th term is a_n , then

$$\begin{aligned} \left| \frac{a_{n+1}}{a_n} \right| &= \left| \frac{k(k-1)\dots(k-n+1)(k-n)x^{n+1}}{(n+1)!} \cdot \frac{n!}{k(k-1)\dots(k-n+1)x^n} \right| \\ &= \frac{|k-n|}{(n+1)} |x| = \frac{|1 - \frac{k}{n}|}{1 + \frac{1}{n}} |x| \xrightarrow{n \rightarrow \infty} |x| \end{aligned}$$

Thus, the Ratio test of the binomial series converges if $|x| < 1$ and diverges for $|x| > 1$.

The traditional notation for the coefficients in the binomial series is

$$\binom{k}{n} = \frac{k(k-1)(k-2)\dots(k-n+1)}{n!}$$

These numbers are called the binomial coefficients.

THE BINOMIAL SERIES

If k is any real number and $|x| < 1$, then

$$(1+x)^k = \sum_{n=0}^{\infty} \binom{k}{n} x^n = 1 + kx + \frac{k(k-1)}{2!} x^2 + \frac{k(k-1)(k-2)}{3!} x^3 + \dots$$

Ex Find the Maclaurin series for $f(x) = \frac{1}{\sqrt{4-x}}$ and its radius of convergence

$$\frac{1}{\sqrt{4-x}} = \frac{1}{\sqrt{4(1-\frac{x}{4})}} = \frac{1}{2\sqrt{1-\frac{x}{4}}} = \frac{1}{2} \left(1 - \frac{x}{4}\right)^{-1/2}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \binom{-1/2}{n} \left(-\frac{x}{4}\right)^n$$

The series converges when $|-x/4| < 1$, that is $|x| < 4$. The radius of convergence is $R=4$

Ex Evaluate $\int e^{-x^2} dx$ as an infinite series

$$e^{-x^2} = \sum_{n=0}^{\infty} \frac{(-x^2)^n}{n!} = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{n!}$$

$$\begin{aligned} \int e^{-x^2} dx &= \sum_{n=0}^{\infty} (-1)^n \int \frac{x^{2n}}{n!} dx \\ &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)n!} + C \end{aligned}$$

The series converges for all x values because the original series e^{-x^2} converges for all x .

$$\int_0^1 e^{-x^2} dx = \left[x - \frac{x^3}{3 \cdot 1!} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \frac{x^9}{9 \cdot 4!} - \dots \right]_0^1$$

$$= 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} - \dots$$

$$\approx 1 - \frac{1}{3} + \frac{1}{10} - \frac{1}{42} + \frac{1}{216} \approx 0.7475$$

the error involved in this approximation is less than

$$\frac{1}{1152} = \frac{1}{1320} < 0.001$$

Ex 7 (9) Approximate the function $f(x) = \sqrt[3]{x}$ by a Taylor polynomial of degree 2 at $a = 8$

$$f(x) = \sqrt[3]{x} = x^{1/3} \quad f(8) = 2$$

$$f'(x) = \frac{1}{3} x^{-2/3} \quad f'(8) = \frac{1}{12}$$

$$f''(x) = -\frac{2}{9} x^{-5/3} \quad f''(8) = -\frac{1}{144}$$

$$f'''(x) = \frac{10}{27} x^{-8/3}$$

$$\left. \begin{array}{l} T_2(x) = f(8) + \frac{f'(8)}{1!} (x-8) + \frac{f''(8)}{2!} (x-8)^2 \\ = 2 + \frac{1}{12} (x-8) - \frac{1}{288} (x-8)^2 \end{array} \right\}$$

The desired approximation is $\sqrt[3]{x} \approx T_2(x) = 2 + \frac{1}{12} (x-8) - \frac{1}{288} (x-8)^2$

(b) How accurate is the approximation when $7 \leq x \leq 9$

Because $x \geq 7$, we have $x^{8/3} \geq 7^{8/3}$

$$\text{and so } f'''(x) = \frac{10}{27} \cdot \frac{1}{x^{8/3}} \leq \frac{10}{27} \cdot \frac{1}{7^{8/3}} < \boxed{0.0021}^M$$

$$|R_2(x)| \leq \frac{M}{3!} |x-8|^3 = \frac{0.0021}{3!} 1^3 = \frac{0.0021}{6} < 0.0004$$

Thus if $7 \leq x \leq 9$, the approximation in part (a)

